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Aircraft maintenance system with no fault found part alerts

Abstract

A method manages aircraft parts. A determination is made as to whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type. The no fault found parameter is how often a no fault found for the part type occurs relative to how often that the part type has been removed and replaced. A determination is made as to whether an average fix time parameter is greater than a fix time threshold for a correct part replacement. The average fix time parameter identifies an average amount of time before a replaced part for the part type was replaced again. An alert is generated in response to the no fault found parameter being greater than the no fault found threshold and the average fix time parameter being greater than fix time threshold.

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Background/Summary

BACKGROUND INFORMATION

1. Field

(1) The present disclosure relates generally to an improved maintenance system for aircraft and in particular, to a method, apparatus, system, and computer program product for managing aircraft parts.

2. Background

(2) Aircraft maintenance involves performing operations to ensure the continued airworthiness of aircraft. Aircraft maintenance can include, for example, inspection, part replacement, overhaul, modifications, and other operations. These operations can include scheduled inspections. The scheduling of these inspections can occur after a period of time, a selected amount of flight time, or a number of aircraft flights.

(3) In addition to scheduled maintenance, computer systems on an aircraft can be used to indicate when maintenance may be needed. For example, a computer system in an aircraft can be a flight management system that is an onboard multipurpose computer system that can automate in-flight tasks, including in-flight management of a flight plan. The flight management system receives data from sensors for different systems in the aircraft.

(4) For example, data can be received from systems such as a propulsion system, an environmental control system, a hydraulic system, a flight control system, and other systems. The flight management system can include aircraft fault logic and performance analysis processes to

determine when a part may need maintenance. A part that is identified as needing maintenance can be removed from the aircraft and replaced with another part.

(5) The removed parts are sent to an inspection facility in which the part is inspected and analyzed to determine what maintenance may be needed on the part. After inspection and performance of any needed maintenance, a part is often returned to a spare parts pool for future use when the maintenance can be performed rather than replacing a part. In some instances, the inspection reveals that no maintenance is needed. This part is marked or labeled as no fault found (NFF). This part is also placed in inventory for reuse in aircraft maintenance.

(6) This process of identifying parts needing maintenance from indications generated by the flight management system in an aircraft can be a time-consuming and expensive process that involves removing a part, replacing a part, performing maintenance, and other operations. Therefore, it would be desirable to have a method and apparatus that takes into account at least some of the issues discussed above such as NFF, as well as other possible issues. Having a method and apparatus that overcomes a technical problem by reducing the amount of maintenance on parts identified by flight management systems can reduce the amount of cost and time needed for maintenance of aircraft.

SUMMARY

(7) An embodiment of the present disclosure provides a method for managing aircraft parts. A computer system determines whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type. The no fault found parameter is how often a no fault found for the part type occurs relative to a number of times that the part type has been removed and replaced. The computer system determines whether an average fix time parameter is greater than a fix time threshold for a correct part replacement. The average fix time parameter identifies an average amount of time before a replaced part for the part type was replaced again. The computer system generates an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than fix time threshold for the correct part replacement.

(8) Another embodiment of the present disclosure provides an aircraft maintenance system comprising a computer system and a maintenance manager in the computer system. The maintenance manager is configured to determine whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type. The no fault found parameter is how often a no fault found for the part type occurs relative to a number of times the part type has been removed and replaced. The maintenance manager is configured to determine whether an average fix time parameter is greater than a fix time threshold for a correct part replacement. The average fix time parameter identifies an average amount of time before a replaced aircraft part for the part type was replaced again. The maintenance manager is configured to generate an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than the fix time threshold for the correct part replacement.

(9) Another embodiment of the present disclosure provides a computer program product for managing aircraft parts. The computer program product comprises a computer readable storage medium having program instructions embodied therewith. The program instructions are executable by a computer system to cause the computer system to perform a method of determining whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type. The no fault found parameter is how often a no fault found for the part type occurs relative to a number of times the part type has been removed and replaced; determining whether an average fix time parameter is greater than a fix time threshold for a correct part replacement, wherein the average fix time parameter identifying an

average amount of time before a replaced part for the part type was replaced again; and generating an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than the fix time threshold for the correct part replacement.

(10) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1 is a pictorial representation of a network of data processing systems in which illustrative embodiments may be implemented;
- (3) FIG. 2 is a block diagram of a maintenance environment in accordance with an illustrative embodiment;
- (4) FIG. 3 is an illustration of a block diagram of a graphical user interface for displaying alerts in accordance with an illustrative embodiment;
- (5) FIG. 4 is an illustration of a graphical user interface displaying alerts for part types in accordance with an illustrative embodiment;
- (6) FIG. 5 is an illustration of a graphical user interface displaying part information for a part type in accordance with an illustrative embodiment;
- (7) FIG. 6 is an illustration of a graph of no fault found related information for a part type in accordance with an illustrative embodiment;
- (8) FIG. 7 is an illustration of a flowchart of a process for generating alerts for part types in accordance with an illustrative embodiment;
- (9) FIG. 8 is an illustration of a flowchart of a process for managing aircraft parts in accordance with an illustrative embodiment;
- (10) FIG. 9 is an illustration of a flowchart of process for generating an alert in accordance with an illustrative embodiment;
- (11) FIG. 10 is an illustration of a flowchart of process for generating an alert in accordance with an illustrative embodiment;
- (12) FIG. 11 is an illustration of a flowchart of process for generating an alert in accordance with an illustrative embodiment;
- (13) FIG. 12 is an illustration of a flowchart of process for generating an alert in accordance with an illustrative embodiment;
- (14) FIG. 13 is an illustration of a flowchart of process for generating an alert in accordance with an illustrative embodiment;
- (15) FIG. 14 is an illustration of a flowchart of process for generating an alert in accordance with an illustrative embodiment;
- (16) FIG. 15 is an illustration of a flowchart of a process for displaying an alert in accordance with an illustrative embodiment;
- (17) FIG. 16 is an illustration of a flowchart of a process for displaying no fault found related information in accordance with an illustrative embodiment;

- (18) FIG. **17** is an illustration of a flowchart of a process for displaying no fault found related information in a graph in accordance with an illustrative embodiment;
- (19) FIG. **18** is an illustration of a flowchart of a process for displaying a no fault found graphical indicator in accordance with an illustrative embodiment;
- (20) FIG. **19** is an illustration of a flowchart of a process for displaying a no fault found graphical indicator in accordance with an illustrative embodiment;
- (21) FIG. **20** is an illustration of a block diagram of a data processing system in accordance with an illustrative embodiment;
- (22) FIG. **21** is an illustration of an aircraft manufacturing and service method in accordance with an illustrative embodiment; and
- (23) FIG. **22** is an illustration of a block diagram of an aircraft in which an illustrative embodiment may be implemented.

DETAILED DESCRIPTION

- (24) The illustrative embodiments recognize and take into account one or more different considerations. For example, the illustrative embodiments recognize and take into account that aircraft parts are expensive to procure and store in inventory. As a result, aircraft fault logic and troubleshooting procedures are designed to discourage unnecessary removal of parts. Even with these aircraft fault logic and troubleshooting procedures, a shop repair or maintenance facility will often report that a component removed for troubleshooting was identified as no fault found (NFF) after testing.
- (25) This component is a no fault found component and can be placed in inventory for a spare parts panel. This process can add costs to airline operations.
- (26) Thus, the illustrative embodiments can provide a method, apparatus, system, and computer program product for managing aircraft parts. An analysis of maintenance data can be performed to identify recommendations based on historical maintenance records. For example, identifying the occurrence of no fault found (NFF) parts can be used to determine whether the frequency of the occurrence for particular part types indicate an issue for further resolution. For example, if the percent of parts of a part type pulled for repair have a no fault found result that is greater than a threshold, an alert can be generated. In some cases, the maintenance procedure may need updating because the part that was pulled and identified as no fault found may not be the part that caused the fault indicator to be generated. In another example, the higher incidence of a no fault found part may indicate that the procedures for evaluating the part may be inaccurate and need updating.
- (27) With reference now to the figures and, in particular, with reference to FIG. **1**, a pictorial representation of a network of data processing systems is depicted in which illustrative embodiments may be implemented. Network data processing system **100** is a network of computers in which the illustrative embodiments may be implemented. Network data processing system **100** contains network **102**, which is the medium used to provide communication links between various devices and computers connected together within network data processing system **100**. Network **102** may include connections, such as wire, wireless communication links, or fiber optic cables.
- (28) In the depicted example, server computer **104** and server computer **106** connect to network **102** along with storage unit **108**. In addition, client devices **110** connect to network **102**.
- (29) As depicted, client devices **110** include airplane **112**, airplane **114**, client computer **116**, maintenance facility **118**, airplane **120**, and airplane **122**. In addition to these depicted examples, client devices **110** can be, for example, computers, workstations, network computers, vehicles, ships, buildings, and other objects or structures that contain data processing systems.
- (30) In this illustrative example, network **102** is the communications media for these network devices. Some or all of client devices **110** may form an Internet of things (IoT) in which these physical devices can connect to network **102** and exchange information with each other over network **102**.
- (31) In the depicted example, server computer **104** provides information, such as boot files,

operating system images, and applications to client devices **110**.

(32) Client devices **110** are clients to server computer **104** in this example. Client devices **110** connect to network **102** utilizing at least one of wired, optical fiber, or wireless connections. Network data processing system **100** can include additional server computers, client computers, aircraft, and other devices not shown.

(33) Program instructions located in network data processing system **100** can be stored on a computer-recordable storage medium and downloaded to a data processing system or other device for use. For example, program instructions can be stored on a computer-recordable storage medium on server computer **104** and downloaded to client devices **110** over network **102** for use on client devices **110**.

(34) In the depicted example, network data processing system **100** is the Internet with network **102** representing a worldwide collection of networks and gateways that use the Transmission Control Protocol/Internet Protocol (TCP/IP) suite of protocols to communicate with one another. At the heart of the Internet is a backbone of high-speed data communication lines between major nodes or host computers consisting of thousands of commercial, governmental, educational, and other computer systems that route data and messages. Of course, network data processing system **100** also may be implemented using a number of different types of networks. For example, network **102** can be comprised of at least one of the Internet, an intranet, a local area network (LAN), a metropolitan area network (MAN), or a wide area network (WAN). FIG. **1** is intended as an example, and not as an architectural limitation for the different illustrative embodiments.

(35) As used herein, “a number of” when used with reference to items, means one or more items. For example, “a number of different types of networks” is one or more different types of networks.

(36) Further, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

(37) For example, without limitation, “at least one of item A, item B, or item C” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

(38) As depicted, maintenance manager **130** is located in server computer **104**. Maintenance manager **130** can manage maintenance for client devices **110** such as airplane **112**, airplane **114**, airplane **120**, and airplane **122**. Further, maintenance manager **130** can also manage parts used to perform maintenance for these aircraft.

(39) In this illustrative example, maintenance manager **130** can receive information from client devices **110**. For example, airplane **112** can send fault indicator **132** to maintenance manager **130**. In this example, both indicator **132** is generated by flight management system and airplane **112**. In response to receiving fault indicator **132**, maintenance manager **130** can schedule maintenance for airplane **112**. This maintenance for airplane **112** can be performed at maintenance facility **118**.

(40) In one illustrative example, a part can be removed and replaced with a replacement part at maintenance facility **118**. In this example, the removed part can be analyzed at maintenance facility **118**.

(41) The part removed from airplane **112** can be analyzed at maintenance facility **118** to determine whether an inconsistency is present that prevents performance of the part at a desired level. This desired level can be based on specifications and tolerances for the part. If the part does not have an inconsistency affecting performance of the part, the part can be labeled or designated as no fault found (NFF). In this case, the part can be placed into inventory for use in maintenance operations.

(42) In this illustrative example, maintenance facility **118** can send this information and other

records regarding maintenance of parts to maintenance manager **130**. This information can be used to manage parts and store historical information about maintenance performed on different aircraft in maintenance database **134**.

(43) In this illustrative example, maintenance manager **130** can analyze maintenance information and maintenance database **134**. For example, maintenance manager **130** can analyze maintenance information to identify part types that have an occurrence of no fault found indicating that the replacement of the part may not be the cause of the fault indicator initiating the removal and replacement of parts for that part type.

(44) In this illustrative example, maintenance manager **130** can generate alerts indicating that additional analysis may be needed for particular part types. These alerts may identify aircraft parts that have been removed unnecessarily in response to the generation of fault indicators by the aircraft. Alternatively, the alerts may indicate that the analysis of the removed aircraft parts may be faulty.

(45) The illustration of network data processing system **100** is an example of one implementation in for an illustrative example. For example, the aircraft depicted in this figure are commercial passenger airplanes. In other illustrative examples, other types of aircraft may be involved including, for example, cargo airplanes, military airplanes, rotorcraft, or other types of aircraft. As another example, the part can be removed and replaced at another location such as a hanger and sent to maintenance facility **118** for analysis. In another illustrative example, the occurrence of fault indicator **132** may result in multiple parts being removed. In some cases, other removed parts may not have been identified as faulty but had been removed and replaced to reduce future maintenance costs.

(46) With reference now to FIG. 2, a block diagram of a maintenance environment is depicted in accordance with an illustrative embodiment. In this illustrative example, maintenance environment **200** includes components that can be implemented in hardware such as the hardware shown in network data processing system **100** in FIG. 1.

(47) In this illustrative example, maintenance system **202** in maintenance environment **200** can operate to manage aircraft parts **229**. In this illustrative example, maintenance system **202** comprises computer system **212** and maintenance manager **214**.

(48) Maintenance manager **214** can be implemented in software, hardware, firmware or a combination thereof. When software is used, the operations performed by maintenance manager **214** can be implemented in program instructions configured to run on hardware, such as a processor unit. When firmware is used, the operations performed by maintenance manager **214** can be implemented in program instructions and data and stored in persistent memory to run on a processor unit. When hardware is employed, the hardware can include circuits that operate to perform the operations in maintenance manager **214**.

(49) In the illustrative examples, the hardware can take a form selected from at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations. With a programmable logic device, the device can be configured to perform the number of operations. The device can be reconfigured at a later time or can be permanently configured to perform the number of operations. Programmable logic devices include, for example, a programmable logic array, a programmable array logic, a field programmable logic array, a field programmable gate array, and other suitable hardware devices. Additionally, the processes can be implemented in organic components integrated with inorganic components and can be comprised entirely of organic components excluding a human being. For example, the processes can be implemented as circuits in organic semiconductors.

(50) Computer system **212** is a physical hardware system and includes one or more data processing systems. When more than one data processing system is present in computer system **212**, those data processing systems are in communication with each other using a communications medium. The

communications medium can be a network. The data processing systems can be selected from at least one of a computer, a server computer, a tablet computer, or some other suitable data processing system.

(51) As depicted, computer system **212** includes a number of processor units **216** that are capable of executing program instructions **218** implementing processes in the illustrative examples. In other words, program instructions **218** are computer readable program instructions.

(52) As used herein, a processor unit in the number of processor units **216** is a hardware device and is comprised of hardware circuits such as those on an integrated circuit that respond to and process instructions and program instructions that operate a computer.

(53) When the number of processor units **216** executes program instructions **218** for a process, the number of processor units **216** can be one or more processor units that are on the same computer or on different computers. In other words, the process can be distributed between processor units **216** on the same or different computers in computer system **212**.

(54) Further, the number of processor units **216** can be of the same type or different type of processor units. For example, the number of processor units **216** can be selected from at least one of a single core processor, a dual-core processor, a multi-processor core, a general-purpose central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or some other type of processor unit.

(55) Maintenance manager **214** can use maintenance data **220** for plurality of aircraft **224** to analyze maintenance performed on plurality of aircraft **224** in which parts were pulled and replaced. In this example, plurality of aircraft **224** can be selected from a group comprising an airplane, a commercial airplane, a cargo plane, a rotorcraft, a tilt wing aircraft, an electrical vertical takeoff and landing vehicle, a personal air vehicle, a military aircraft, or other types of aircraft.

(56) Maintenance data **220** is located in maintenance database **221** in this example. Maintenance data **220** relating to maintenance performed on the plurality of aircraft **224** can be obtained from a number of sources **223**. For example, maintenance data **220** can be received from at least one of a shop repair facility, a maintenance facility, an airline, or other sources.

(57) In this illustrative example, maintenance data **220** includes information about part types, aircraft parts, fault indicators, dates when fault occurred, maintenance performed, parts removed, parts replaced, and other information relating to the maintenance for plurality of aircraft **224**. For example, maintenance data **220** can include shop data from a shop or repair facility. The shop data can identify a part type, whether the part type is a no fault found (NFF) part. The maintenance data can also include parts replaced, tests performed, notes, and other information in this example, maintenance manager **214** analyzes maintenance data **220** for part type **228** from plurality of aircraft **224**. In the illustrative example, plurality of aircraft **224** can be aircraft of the same model. Plurality of aircraft **224** can also include variances of that model.

(58) In this illustrative example, aircraft parts **229** of part type **228** are parts that can be used in a location in the aircraft. For example, aircraft parts **229** of part type **228** include various parts in aircraft parts **229** that have the same fit, form, and function but may have different part numbers.

(59) In one illustrative example, maintenance manager **214** determines whether no fault found parameter **226** for part type **228** is greater than no fault found threshold **230** for an acceptable level for no fault found occurrences **232** for part type **228**. No fault found parameter **226** is how often no fault found **234** for part type **228** occurs relative to a number of times that part type **228** has been removed and replaced.

(60) In other words, no fault found parameter **226** is a ratio of the number of times aircraft parts **229** for part type **228** were identified as no fault found to the number of times aircraft parts **229** for part type **228** were removed and replaced. This ratio can be expressed as a percentage in some illustrative examples.

(61) In this depicted example, no fault found threshold **230**, is selected to indicate when no fault found parameter **226** is considered to be too high to be related to the problem for which

maintenance was performed on plurality of aircraft **224**. In this example, part type **228** can have no fault found parameter **226** that is too high when the number of parts of part type **228** is great enough that replacing part type **228** is not considered to be the component or part that fixes the problem for which maintenance was performed. As the percentage or ratio of no fault found parameter **226** increases, it becomes more likely that the replacement of an aircraft part of part type **228** is not an actual fix for the problem.

(62) Alternatively, as no fault found parameter **226** increases for part type **228**, it can be more likely that the testing procedures for part type **228** may be faulty or unable to detect issues with aircraft parts **229** of part type **228**. As a result, the testing of aircraft parts **229** of part type **228** pulled from the plurality of aircraft **224** may have fixed the problem but inconsistencies or other issues with pulled aircraft parts were not identified in the testing of those aircraft parts. In this illustrative example, the threshold can be selected based on analyzing statistical distributions. The threshold can be a percentage that is statistically out of balance from what is considered to be normal for the part.

(63) Maintenance manager **214** determines whether average fix time parameter **236** is greater than fix time threshold **238** for correct part replacement **241**. In this example, average fix time parameter **236** is the average amount of time before the replacement aircraft part needs to be replaced again for part type **228**. In these examples, average fix time parameter **236** identifies the average amount of time before a replacement part of part type **228** that fixes a problem needs to be replaced again.

(64) In this example, fix time is the amount of time that a particular part in an aircraft lasts before needing to be replaced again in that aircraft. The replacement can be initiated by a fault indicator generated in the aircraft in which the part is located.

(65) The fix time for the different parts in plurality of aircraft **224** is averaged to obtain average fix time parameter **236** for part type **228**. In these illustrative examples, the average amount of time can be measured in a number of different ways. For example, the average amount of time can be measured in hours, days, months, or some other period of time. When the amount of time is measured in days, average fix time parameter **236** can be referred to as an average fix days parameter.

(66) As the average amount of time increases with how long aircraft part fixes a problem, it is more likely that the replacement of that aircraft part was a proper fix for the issue resulting from performing the maintenance that pulled and replaced the aircraft part. In this example, fix time threshold **238** is selected to have a value such that confidence is present that the replacement part a resulted in a fix and that another replacement of the aircraft part is not needed within what is considered an expected time for replacing that part. This threshold is intended to make sure the fix for part type **228** does not last for too short of a period of time relative to the amount of time the part type is expected to last. In this example, expected time can be obtained from aircraft part manufacturers. This threshold can be selected to remove data when the replacement did not fix the problem for a time that is considered sufficient count as an issue for no fault found parts.

(67) In this illustrative example, maintenance manager **214** can use these two parameters, no fault found parameter **226**, an average fix time parameter **236**, to determine whether to generate alert **240**. For example, maintenance manager **214** generates alert in response to no fault found parameter **226** for part type **228** being greater than no fault found threshold **230** for the acceptable level for no fault found occurrences **232** for part type **228** and average fix time parameter **236** being greater than fix time threshold **238** for correct part replacement.

(68) In this example, the alert is generated when no fault found parameter **226** is higher than the threshold for this parameter and average fix time parameter **236** is also higher than the threshold for the parameter. When these two parameters are greater than the thresholds, the parameters are considered to be high.

(69) In this example, when the no fault found percentage is high and the average fix time is also high, further investigation is needed. Further investigation is warranted because no fault found

parameter **226** indicates that the percentage of aircraft parts **229** of part type **228** that are no fault found parts are higher than expected for that part to fix the problem and average fix time parameter **236** indicates that replacing aircraft parts **229** of part type **228** fixes the problem.

(70) This situation can be present when at least one of replacement of aircraft parts **229** of part type **228** may not be a correct fix or the testing procedures resulting in finding no fault found **234** for part type **228** may not actually identify an issue for part type **228**.

(71) In these illustrative examples, maintenance manager **214** can take into account the additional factors in generating alert **240**. For example, maintenance manager **214** can determine whether fix count **242** identifying a number of times removal and replacement of aircraft parts **229** of part type **228** fixed a problem is greater than median fix count **244** multiplied with fix count median threshold **246**.

(72) In this example, fix count **242** is the number of times that the replacement of a part for part type **228** resulted in fixing the problem. Fix count **242** can be used to track how well replacing an aircraft part fixes a problem at the aircraft or airplane level through identifying how long problem went away.

(73) Each time the part is replaced for a particular airplane, if the fault being repaired stops occurring for a certain number of days, that replacement increments fix count **242** for that aircraft part in that airplane. For example, if a satellite unit for location X is replaced in aircraft A, and the fault being repaired stops occurring for a certain number of days, the fix count is one. The fix count increases to two the next time the satellite unit is replaced for location X in aircraft A, and the fault does not reoccur for a certain number of days.

(74) In this example, the median fix count **244** is a median value for all of the fix counts all of the different parts in the plurality of aircraft **224** for part type **228**. The median is the value in the middle of the data set which comprises fix count **242** for all of plurality of aircraft **224**.

(75) With taking into account fix count **242**, maintenance manager **214** generates alert **240** in response to no fault found parameter **226** for part type **228** being greater than no fault found threshold **230** for the acceptable level for no fault found occurrences **232** for part type **228**; average fix time parameter **236** being greater than fix time threshold **238** for correct part replacement **241**; and fix count **242** identifying the number of times removal and replacement of parts for the part type fixed the problem being greater than median fix count **244** multiplied with fix count median threshold **246**.

(76) As another example, maintenance manager **214** can determine whether shop count **248** identifying a number of times aircraft parts **229** of part type **228** have been removed and replaced is greater than median shop count **250** multiplied with shop count median threshold **252**. In this illustrative example, shop count **248** is a number of times aircraft parts **229** of part type **228** have been removed from plurality of aircraft **224** and sent for inspection or testing.

(77) More specifically, shop count **248** can be the number of times a particular part for an aircraft in plurality of aircraft **224** has been removed. In other words, this parameter can identify the number of times a part has been removed and replaced in an aircraft. Median shop count **250** is the median value for shop count **248** for all of plurality of aircraft **224** that use part type **228**.

(78) With taking into account shop count **248**, maintenance manager **214** generates alert **240** in response to no fault found parameter **226** for part type **228** being greater than no fault found threshold **230** for the acceptable level for no fault found occurrences **232** for part type **228**; average fix time parameter **236** being greater than fix time threshold **238** for correct part replacement **241**; and shop count **248** identifying a number of times aircraft parts **229** for part type **228** have been removed and replaced being greater than median shop count **250** multiplied with shop count median threshold **252**.

(79) In this illustrative example, fix count **242** and shop count **248** can be used to determine whether to consider part type **228** with respect to generating alert **240**. These two parameters individually or collectively can be used to determine whether sufficient data is present to consider

generating alert **240**. In other words, part type **228** can be removed from consideration when the sample size is not sufficiently large to be meaningful. Additionally, both fix count **242** and shop count **248** can be taken into account at the same time in other illustrative examples.

(80) In this illustrative example, alert **240** can be displayed to operator **253** using human machine interface (HMI) **254** in computer system **212**. The display of alert **240** can be with other information in maintenance data **220**. Further, alert **240** can be specifically displayed in association with information about part type **228** and aircraft parts **229** that are part of part type **228**.

(81) In this illustrative example, human machine interface **254** comprises display system **256** and input system **259**. Operator **253** interacts with maintenance system **202** using human machine interface **254**.

(82) In this illustrative example, display system **256** is a physical hardware system and includes one or more display devices on which graphical user interface **258** can be displayed. The display devices can include at least one of a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a computer monitor, a projector, a flat panel display, a heads-up display (HUD), a head-mounted display (HMD), smart glasses, augmented reality glasses, or some other suitable device that can output information for the visual presentation of information.

(83) Operator **253** is a person that can interact with graphical user interface **258** through user input **260** generated by input system **259** for computer system **212**. Input system **259** is a physical hardware system and can be selected from at least one of a mouse, a keyboard, a touch pad, a trackball, a touchscreen, a stylus, a motion sensing input device, a gesture detection device, a data glove, a cyber glove a haptic feedback device, or some other suitable type of input device.

(84) Turning next to FIG. **3**, an illustration of a block diagram of a graphical user interface for displaying alerts is depicted in accordance with an illustrative embodiment. In the illustrative examples, the same reference numeral may be used in more than one figure. This reuse of a reference numeral in different figures represents the same element in the different figures.

(85) An example of an implementation for graphical user interface **258** in FIG. **2** is shown in this figure. In this example, maintenance manager **214** displays part types **300** in graphical user interface **258**. Further, maintenance manager **214** can display alert graphical indicator **302** in association with part type **228** in part types **300**. Alert graphical indicator **302** is a graphical representation of alert **240** in FIG. **2** and visually indicates alert **240** in graphical user interface **258**. In this example, the display of alert graphical indicator **302** in association with part type **228** means that the display of alert graphical indicator **302** is such that attention is drawn to part type **228** in graphical user interface **258**.

(86) In the illustrative example, a graphical indicator can include at least one of an icon, a pictogram, an ideogram, a graphic, an image, text, animation, bolding, a color, a line, an arrow, or other suitable graphic. In other words, the graphical indicator can be a single graphical element or combination of different graphical elements.

(87) In this illustrative example, maintenance manager **214** can display table **305** in graphical user interface **258**. With this example, part types **300** can be displayed in rows **306** in graphical user interface **258**. No fault found related information **308** can be displayed in columns **310**. No fault found related information **308** can include at least one of a shop count, a no fault found count, average fix time, no fault found percent, or other information.

(88) As another example, maintenance manager **214** can display graph **320** in graphical user interface **258**. Maintenance manager **214** can display at least one a shop count, a no fault found count, average fix time, or no fault found percent over selected periods of time in graph **320** in response to a selection of the part type.

(89) As another example, maintenance manager **214** can display no fault found percent **330** for part type **304** in the graphical user interface **258** using no fault found graphical indicator **332**. In this example, no fault found graphical indicator **332** comprises graphical elements that include number **334** representing no fault found percent **330** and trend line **336** representing a trend for the no fault

found percent **330**.

(90) Maintenance manager **214** can also enable drilling down to more detailed information. For example, maintenance manager **214** can display percent change **338** in the trend over a selected period of time in graphical user interface **258** in response to a selection of the no fault found graphical indicator **332**.

(91) In this illustrative example, graphical user interface **258** provides a visualization of information through the use of graphical elements in the form of graphical indicators. These graphical indicators can be displayed in association with information such as information about parts displayed in the graphical user interface. These graphical indicators can enable a human operator to more quickly comprehend information about parts displayed in the graphical user interface in contrast merely displaying text in tables.

(92) In these illustrative examples, the graphical indicators used can provide information about no fault found parts such alerts that part types for the no fault found parts may need further analysis. Further, the alerts are generated in a manner to indicate that the no fault found parts pulled from the aircraft may not have contributed to fixing the problem. For example, employing parts to perform maintenance in response to a fault indicator generated by an aircraft, more than one part can be pulled and replaced. Some of these parts can be pulled and replaced to perform maintenance specifically for the fall indicator. In other parts can be pulled to perform other maintenance.

(93) Additionally, the alerts can also indicate that the replacement of the parts for the part type fix the problem even though these parts have been identified as no fault found. Further, a percentage of no fault found parts, trendlines, and other information can be graphically displayed in a manner that is more easily received and understood by the human operator as compared to merely displaying text. The graphical indicators in the illustrative examples provide indications of not just information about part types but identify situations in which further examination of part types is needed.

(94) Illustration of the different graphical elements for displaying maintenance information and alerts in graphical user interface **258** in FIG. **2** is intended illustrate alerts that can be displayed for part types. This illustration is not meant to limit the manner in which other illustrative examples can be implemented. In other examples, other types of information can be displayed in addition to or in place of the information affected. For example, aircraft parts for a particular part type can be displayed for that particular part type. In this example, aircraft parts for parts that have the same number. In some cases, parts in the same part type can have different part numbers representing different versions of a part. Further, no fault found related information **308** can also be displayed for the particular parts in a part type.

(95) In still other illustrative examples, other information such as recommendations on actions to be taken with respect to parts having alerts indicated by alert graphical indicator **302** or no fault found graphical indicator **332** can be made in graphical user interface **258** by maintenance manager **214**.

(96) For example, the recommendation can be to analyze at least one of the current maintenance procedure or the current testing procedure used for part type **228**. Actions that can be taken can result in at least one of a new maintenance procedure or a new testing procedure for part type **228**. The new maintenance procedure can be an entirely new procedure or an update or modification to the current maintenance procedure for part type **228**. The new testing procedure also can be an entirely new procedure or an update or modification to the current testing procedure for part type **228**.

(97) For example, actions can be taken to update maintenance procedures as to what parts to pull and replace. Further, actions can also be taken to update testing procedures for parts that have been pulled for testing.

(98) With at least one of a new maintenance procedure or a new testing procedure for part type **228**, a reduction can occur in the number of parts that are pulled and replaced. This reduction can occur

through improved identification of which parts should be pulled or replaced based on the analysis of no fault found parts for which alerts have been generated. Additionally, parts that actually need maintenance can be identified rather than being returned to a pool of parts for reuse. As a result, less time and expense can occur in performing maintenance for plurality of aircraft **224**.

(99) In one illustrative example, one or more technical solutions are present that overcome a technical problem with managing aircraft maintenance in a manner that takes into account occurrences of no fault found parts. As a result, one or more solutions can enable identifying when a particular part type has occurrences of no fault found for parts in the part type or are sufficient to warrant further analysis. In the illustrative examples, an alert is generated when occurrences of no fault found for parts in a part type exceed thresholds identified for generating alerts. Further, thresholds also can be used to avoid the generation of an alert when insufficient data is present for a particular part type.

(100) Computer system **212** can be configured to perform at least one of the steps, operations, or actions described in the different illustrative examples using software, hardware, firmware or a combination thereof. As a result, computer system **212** operates as a special purpose computer system in which maintenance manager **214** in computer system **212** enables managing aircraft parts in which no fault found parts are present. In particular, maintenance manager **214** transforms computer system **212** into a special purpose computer system as compared to currently available general computer systems that do not have maintenance manager **214**.

(101) In the illustrative example, the use of maintenance manager **214** in computer system **212** integrates processes into a practical application for managing aircraft parts as compared to current part management systems. In other words, maintenance manager **214** in computer system **212** is directed to a practical application of processes integrated into maintenance manager in computer system **212** that determines a no fault found parameter and an average fix time parameter. These parameters are used to determine whether an alert should be generated with respect to no fault found occurrences for that part type. These parameters can be used to determine whether the no fault found parameter and the average fix time parameter, or both, exceed thresholds.

(102) In this illustrative example, the threshold for the no fault found parameter is a level that indicates that parts being removed of the part type do not need maintenance. The threshold for the fix time parameter is at a level that indicates that parts being removed of the part type fix the issue for the part type. When both of these parameters exceed thresholds, an indication is present that parts of the part type by being pulled and replaced that are good parts and that pulling and replacing the parts fixes the problem. When both of these conditions are present, the maintenance that pulls and replaces the parts of the part type may not be for a part that needs to be fixed or repaired. Alternatively, the testing procedure for parts of the part type may be incorrect and may need revisions.

(103) The illustration of the different components in maintenance environment **200** in in FIGS. 2-3 is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

(104) For example, maintenance manager **214** can also schedule maintenance for plurality of aircraft **224** in addition to analyzing maintenance data **220**. In other illustrative examples, part type **228** can be for parts for vehicles than aircraft parts **229**. For example, part type **228** can be for parts for a vehicle selected from a group comprising a mobile platform, a surface ship, a tank, a personnel carrier, a train, a spacecraft, a submarine, a bus, an automobile, and other suitable types of vehicles.

(105) Turning now to FIG. 4, an illustration of a graphical user interface displaying alerts for part types is depicted in accordance with an illustrative embodiment. In this illustrative example,

graphical user interface **400** is an example of one implementation for graphical user interface **258** in FIG. 2 and FIG. 3.

(106) As depicted, table **402** is displayed within graphical user interface **400**. Table **402** is an example of an implementation for table **305** in FIG. 3.

(107) As depicted, table **402** comprises rows **404** and columns **406**. As depicted, part types are illustrated for multiple aircraft.

(108) In this example, alerts are graphically displayed in alert column **408**. When an alert regarding a no fault found is generated, this alert is displayed in alert column **408** for a particular part type using an alert graphical indicator. For example, graphical alert indicator **410** is displayed in alert column **408** in row **412** for the part type, control card (center).

(109) In this example, no fault found related information is also displayed in the columns. For example, equipment number column **420** identifies can identify a part type. The equipment number can be the same for a part type that can be used in different locations.

(110) In this example, name column **401** identifies the part type. Shop count column **422** identifies the number of parts removed and replaced for a part type. NFF count column **424** identifies the number of parts that were identified as no fault found after testing, and fix count column **426** identifies how many times the removal and replacement of the part fixed or resolved the problem. Average fix days column **428** is the average number of days that parts that were removed and replaced fix the problem.

(111) Further, table **402** also includes no fault found percent column **432**. This column includes no fault found graphical indicators that indicate the percent of parts in a part type that were found to be no fault found parts. Further, these graphical indicators also indicate a trend for the percentage.

(112) For example, no fault found graphical indicator **440** has graphical elements comprising number **442** identifying the no fault found percentage as 85% and trendline **444** indicating an upward trend in the percent of no fault found parts. In another example, no fault found graphical indicator **446** has graphical elements comprising number **448** identifying the no fault found percentage as 37% and trendline **450** indicating a downward trend in the percent of no fault found parts.

(113) In yet another illustrative example, no fault found graphical indicator **452** has graphical elements comprising number **454** identifying the no fault found percentage as 96% and trendline **456** indicating a downward trend in the percent of no fault found parts. In absence of a graphical indicator in no fault found percent column **432** indicates that insufficient data is present to provide accurate information.

(114) With reference to FIG. 5, an illustration of a graphical user interface displaying part information for a part type is depicted in accordance with an illustrative embodiment. In this illustrative example, table **500** is displayed in graphical user interface **502**. Graphical user interface **502** is an example of an implementation for graphical user interface **258** in FIG. 2 and FIG. 3. Table **500** is an example of an implementation for table **305** in FIG. 3.

(115) In this example, table **500** comprises rows **504** and columns **506**. As depicted, part types are illustrated for multiple aircraft.

(116) In this example, graphical alert indicator **507** is displayed in alert column **508** for an alert for the part type in row **510**. The part type in this example is a left satellite data unit. In this example, the part type is identified in name column **501**.

(117) In this example, no fault found related information is displayed in equipment number column **520**, shop count column **522**, NFF count column **524**, fix count column **526**, and average fix days column **528**. Further, no fault found percent column **532** has no fault found graphical indicators for part types that have sufficient information for analysis.

(118) In this illustrative example, additional information about parts in a part type can be displayed in response to a selection of the part-time. As depicted, the part type left satellite data unit in row **510** has been selected by a user input. In response to the selection of this part type, window **540** is

displayed with information about the parts in this part type.

(119) As depicted, window **540** has row **542** and row **544** corresponding to parts with specific part numbers in part number column **546**. In this example, row **542** represents first parts with part number xxxx-yyyyy-**103** and row **542** is for second parts with part number xxxx-yyyyy-**102**. In this example, these parts are different versions of each other.

(120) Additionally, window **540** also has the shop count for parts in shop count column **560**, the NFF count for the parts in NFF count column **562**, and the NFF percent in NFF percent column **564**.

(121) With reference next to FIG. **6**, an illustration of a graph of no fault found related information for a part type is depicted in accordance with an illustrative embodiment. In this illustrative example, graph **600** is an example of graph **320** and FIG. **3**. In this depicted example, graph **600** is a bar graph identifying no fault found related information over 12 years for parts in a part type that is a satellite antenna. Graph **600** can be displayed in response to a selection of a satellite antenna part type in a table such as table **402** in FIG. **4**.

(122) In this example, y-axis **601** is occurrences of a no fault found, and y-axis **603** is no fault found percentage. X-axis **605** is years.

(123) The no fault found information graphically displayed in graph **600** includes no fault found occurrences **602**, fault found occurrences **604**, NFF percent **606**, and NFF percent trend **608**.

(124) With reference next to FIG. **7**, an illustration of a flowchart of a process for generating alerts for part types is depicted in accordance with an illustrative embodiment. The process in FIG. **7** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that are run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in maintenance manager **214** in computer system **212** in FIG. **2**.

(125) As depicted, the process begins by pulling shop data for analysis (operation **700**). The process groups the shop data by part type (operation **702**). The process selects a part type for processing (operation **704**). In operation **704**, the shop data for the selected part type is identified for use in determining no fault found information about the part type selected for processing.

(126) The process calculates the shop count of parts in the part type (operation **706**). In operation **706**, the shop count is tracked for each instance of the part type in different aircraft. For example, a shop count is tracked for a part of the part type in an aircraft. Each time that part in the aircraft is removed and sent for testing, the shop count for that part is incremented. Thus, the shop count is present for each part of the part type in the aircraft.

(127) The process calculates a no fault found (NFF) percent for the part type (operation **708**). The process calculates a fix count for each part in the part type removed from a number of aircraft (operation **710**). In operation **710**, the fix count is performed on a per aircraft basis. For example, the number of times that the antenna is replaced on aircraft is the fix count for that part. In other words, the part type can be tracked based on the replacement of instances of aircraft parts of that part type in a particular aircraft. The particular aircraft identified using different identifiers such as a tail number.

(128) The process calculates an average fix days for each part in the part type removed from the number of aircraft (operation **712**).

(129) The process calculates a median shop count for the part type (operation **714**). The process calculates a median fix count for each part in the part type removed from the number of aircraft (operation **716**).

(130) The process determines whether the no fault found percentage (NFF %) is greater than a no fault found threshold (operation **718**). If the no fault found percentage is greater than the no fault found threshold, the process determines whether the average fix days is greater than a fix days threshold (operation **720**).

(131) If the average fix days is greater than the fix days threshold, the process determines whether

the fix count is greater than the median fix count multiplied by the median percentile threshold (operation **722**).

(132) If the fix count is greater than the median fix count multiplied by the fix count median percentile threshold, the process determines whether the shop count is greater than the median shop count multiplied by the shop count median percentile threshold (operation **724**).

(133) If the shop count is greater than the median shop count multiplied by the shop count median percentile threshold, the process sets an alert for the part type (operation **726**). The process then determines whether another part type is present for processing (operation **728**). If another part type is not present, the process terminates. On the other hand, if another part type is present for processing, the process returns to operation **704** as described above.

(134) With reference again to operation **718**, if the no fault found percentage is not greater than the no fault found threshold, the process proceeds operation **728** as described above. Turning back to operation **720**, if the average fix days is not greater than the fix days threshold, the process proceeds operation **728** as described above. With reference again to operation **722**, if the fix count is not greater than the median fix count multiplied by the median percentile threshold, the process also proceeds to operation **728**. In a similar fashion with operation **724**, if the shop count is not greater than the median fix count multiplied by the fix count median percentile threshold, the process proceeds operation **728**.

(135) Turning next to FIG. **8**, an illustration of a flowchart of a process for managing aircraft parts is depicted in accordance with an illustrative embodiment. The process in FIG. **8** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that are run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in maintenance manager **214** in computer system **212** in FIG. **2**.

(136) The process begins by determining whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type (operation **800**). In operation **800**, the no fault found parameter is how often a no fault found for the part type occurs relative to a number of times that the part type has been removed and replaced. The no fault found parameter can take a number of different forms. For example, the no fault found parameter can be a number, a percentage, or some other form. When the no fault found parameter is a percentage, the no fault found parameter can be referred to as a no fault found percentage.

(137) The process determines whether an average fix time parameter is greater than a fix time threshold for a correct part replacement (operation **802**). In operation **802**, the average fix time parameter identifies an average amount of time before a replaced part for the part type was replaced again. In one illustrative example, the time to be measured in days. The average fix time parameter can also be referred to as average fix time.

(138) The process generates an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than the fix time threshold for the correct part replacement (operation **804**). The process terminates thereafter.

(139) With reference next to FIG. **9**, an illustration of a flowchart of process for generating an alert is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **9** is an example of an additional operation that can be performed with the operations in FIG. **8**.

(140) The process determines whether a fix count identifying a number of times removal and replacement of the aircraft parts for the part type fixed a problem is greater than a median fix count multiplied with a fix count median threshold (operation **900**). The process terminates thereafter.

(141) Turning to FIG. **10**, an illustration of a flowchart of process for generating an alert is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **9** is an example of an implementation of operation **804** in FIG. **8** when operation **900** in FIG. **9** is used.

(142) The process generates the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for a correct part replacement; and the fix count identifying the number of times removal and replacement of aircraft parts for the part type fixed the problem being greater than the median fix count multiplied with the fix count median threshold (operation **1000**). The process terminates thereafter.

(143) With reference next to FIG. **11**, an illustration of a flowchart of process for generating an alert is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **11** is an example of an additional operation that can be performed with the operations in FIG. **8**.

(144) The process determines whether a shop count identifying a number of times the aircraft parts for the part type have been removed and replaced is greater than a median shop count multiplied with a shop count median threshold (operation **1100**). The process terminates thereafter.

(145) Turning to FIG. **12**, an illustration of a flowchart of process for generating an alert is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **12** is an example of an implementation of operation **804** in FIG. **8** when operation **1100** in FIG. **11** is used.

(146) The process generates the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for the correct part replacement; and shop count identifying a number of times aircraft parts for the part type have been removed and replaced being greater than a median shop count multiplied with a shop count median threshold (operation **1200**). The process terminates thereafter.

(147) With reference next to FIG. **13**, an illustration of a flowchart of process for generating an alert is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **13** is an example of an additional operation that can be performed with the operations in FIG. **8**.

(148) The process begins by determining whether a fix count identifying a number of times removal and replacement of the aircraft parts for the part type fixed a problem is greater than a median fix count multiplied with a fix count median threshold (operation **1300**). The process determines whether a shop count identifying a number of times aircraft parts for the part type have been removed and replaced is greater than a median shop count multiplied with a shop count median threshold (operation **1302**). The process terminates thereafter.

(149) Turning to FIG. **14**, an illustration of a flowchart of process for generating an alert is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **14** is an example of an implementation of operation **804** in FIG. **8** when operation **1300** and operation **1302** in FIG. **13** are used.

(150) The process generates the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for the correct part replacement; the fix count identifying the number of times removal and replacement of aircraft parts for the part type fixed the problem is greater than the median fix count multiplied with the fix count median threshold; and the shop count identifying the number of times aircraft parts for the part type have been removed and replaced being greater than the median shop count multiplied with a shop count median threshold (operation **1400**). The process terminates thereafter.

(151) With reference now to FIG. **15**, an illustration of a flowchart of a process for displaying an alert is depicted in accordance with an illustrative embodiment. The process in FIG. **15** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that are run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in maintenance manager **214** in computer system **212** in FIG. **2**.

(152) The process begins by displaying part types in a graphical user interface on a display system in the computer system (operation **1500**). The process displays an alert graphical indicator in association with the part type in the graphical user interface (operation **1502**). The process terminates thereafter. In operation **1502**, the alert graphic indicator visually indicates the alert in the graphical user interface.

(153) In FIG. **16**, an illustration of a flowchart of a process for displaying no fault found related information is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **16** is an example of additional operations that can be performed with the operations in FIG. **15**.

(154) The process begins by displaying the part types in rows in the graphical user interface (operation **1600**). The process displays at least one of a shop count, a no fault found count, average fix time, or no fault found percent in columns in the graphical user interface (operation **1602**). The process terminates thereafter. In operation **1602**, the shop count, the no fault found count, the average fix time, and the no fault found percent are examples of no fault found related information.

(155) Turning now to FIG. **17**, an illustration of a flowchart of a process for displaying no fault found related information in a graph is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **17** is an example of additional operations that can be performed with the operations in FIG. **15**.

(156) The process displays a graph with at least one of a shop count, a no fault found count, average fix time, or a no fault found percent over selected periods of time in response to a selection of the part type (operation **1700**). The process terminates thereafter. In operation **1700**, the shop count, the no fault found count, the average fix time, and the no fault found percent are examples of no fault found related information.

(157) With reference next to FIG. **18**, an illustration of a flowchart of a process for displaying a no fault found graphical indicator is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **18** is an example of additional operations that can be performed with the operations in FIG. **15**.

(158) The process displays a no fault found percent in the graphical user interface using a no fault found graphical indicator comprising a number representing the no fault found percent and a trend line representing a trend for the no fault found percentage (operation **1800**). The process terminates thereafter.

(159) Next in FIG. **19**, an illustration of a flowchart of a process for displaying a no fault found graphical indicator is depicted in accordance with an illustrative embodiment. The process illustrated in FIG. **19** is an example of additional operations that can be performed with the operations in FIG. **15**.

(160) The process displays a percent change in the trend over a selected period of time in the graphical user interface in response to a selection of the no fault found graphical indicator (operation **1900**). The process terminates thereafter.

(161) The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams can represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks can be implemented as program instructions, hardware, or a combination of the program instructions and hardware. When implemented in hardware, the hardware can, for example, take the form of integrated circuits that are manufactured or configured to perform one or more operations in the flowcharts or block diagrams. When implemented as a combination of program instructions and hardware, the implementation may take the form of firmware. Each block in the flowcharts or the block diagrams can be implemented using special purpose hardware systems that perform the different operations or combinations of special purpose hardware and program instructions run by the special purpose hardware.

(162) In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram.

(163) For example, operation **722** and operation **724** in FIG. 7 are used to remove part types from consideration when insufficient data is present. In some illustrative examples, these determinations can be optional determinations.

(164) Turning now to FIG. 20, a block diagram of a data processing system is depicted in accordance with an illustrative embodiment. Data processing system **2000** can be used to implement server computer **104**, server computer **106**, client devices **110**, in FIG. 1. Data processing system **2000** can also be used to implement computer system **212** in FIG. 2.

(165) In this illustrative example, data processing system **2000** includes communications framework **2002**, which provides communications between processor unit **2004**, memory **2006**, persistent storage **2008**, communications unit **2010**, input/output (I/O) unit **2012**, and display **2014**. In this example, communications framework **2002** takes the form of a bus system.

(166) Processor unit **2004** serves to execute instructions for software that can be loaded into memory **2006**. Processor unit **2004** includes one or more processors. For example, processor unit **2004** can be selected from at least one of a multicore processor, a central processing unit (CPU), a graphics processing unit (GPU), a physics processing unit (PPU), a digital signal processor (DSP), a network processor, or some other suitable type of processor. Further, processor unit **2004** can may be implemented using one or more heterogeneous processor systems in which a main processor is present with secondary processors on a single chip. As another illustrative example, processor unit **2004** can be a symmetric multi-processor system containing multiple processors of the same type on a single chip.

(167) Memory **2006** and persistent storage **2008** are examples of storage devices **2016**. A storage device is any piece of hardware that is capable of storing information, such as, for example, without limitation, at least one of data, program instructions in functional form, or other suitable information either on a temporary basis, a permanent basis, or both on a temporary basis and a permanent basis. Storage devices **2016** may also be referred to as computer-readable storage devices in these illustrative examples. Memory **2006**, in these examples, can be, for example, a random-access memory or any other suitable volatile or non-volatile storage device. Persistent storage **2008** may take various forms, depending on the particular implementation.

(168) For example, persistent storage **2008** may contain one or more components or devices. For example, persistent storage **2008** can be a hard drive, a solid-state drive (SSD), a flash memory, a rewritable optical disk, a rewritable magnetic tape, or some combination of the above. The media used by persistent storage **2008** also can be removable. For example, a removable hard drive can be used for persistent storage **2008**.

(169) Communications unit **2010**, in these illustrative examples, provides for communications with other data processing systems or devices. In these illustrative examples, communications unit **2010** is a network interface card.

(170) Input/output unit **2012** allows for input and output of data with other devices that can be connected to data processing system **2000**. For example, input/output unit **2012** may provide a connection for user input through at least one of a keyboard, a mouse, or some other suitable input device. Further, input/output unit **2012** may send output to a printer. Display **2014** provides a mechanism to display information to a user.

(171) Instructions for at least one of the operating system, applications, or programs can be located in storage devices **2016**, which are in communication with processor unit **2004** through communications framework **2002**. The processes of the different embodiments can be performed by processor unit **2004** using computer-implemented instructions, which may be located in a

memory, such as memory **2006**.

(172) These instructions are referred to as program instructions, computer usable program instructions, or computer-readable program instructions that can be read and executed by a processor in processor unit **2004**. The program instructions in the different embodiments can be embodied on different physical or computer-readable storage media, such as memory **2006** or persistent storage **2008**.

(173) Program instructions **2018** are located in a functional form on computer-readable media **2020** that is selectively removable and can be loaded onto or transferred to data processing system **2000** for execution by processor unit **2004**. Program instructions **2018** and computer-readable media **2020** form computer program product **2022** in these illustrative examples. In the illustrative example, computer-readable media **2020** is computer-readable storage media **2024**.

(174) Computer-readable storage media **2024** is a physical or tangible storage device used to store program instructions **2018** rather than a medium that propagates or transmits program instructions **2018**. Computer readable storage media **2024** may be at least one of an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or other physical storage medium. Some known types of storage devices that include these mediums include: a diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device, such as punch cards or pits/lands formed in a major surface of a disc, or any suitable combination thereof.

(175) Computer-readable storage media **2024**, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as at least one of radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, or other transmission media.

(176) Further, data can be moved at some occasional points in time during normal operations of a storage device. These normal operations include access, de-fragmentation or garbage collection. However, these operations do not render the storage device as transitory because the data is not transitory while the data is stored in computer-readable storage media **2024**.

(177) Alternatively, program instructions **2018** can be transferred to data processing system **2000** using a computer-readable signal media. The computer-readable signal media are signals and can be, for example, a propagated data signal containing program instructions **2018**. For example, the computer-readable signal media can be at least one of an electromagnetic signal, an optical signal, or any other suitable type of signal. These signals can be transmitted over connections, such as wireless connections, optical fiber cable, coaxial cable, a wire, or any other suitable type of connection.

(178) Further, as used herein, “computer-readable media **2020**” can be singular or plural. For example, program instructions **2018** can be located in computer-readable media **2020** in the form of a single storage device or system. In another example, program instructions **2018** can be located in computer-readable media **2020** that is distributed in multiple data processing systems. In other words, some instructions in program instructions **2018** can be located in one data processing system while other instructions in program instructions **2018** can be located in one data processing system. For example, a portion of program instructions **2018** can be located in computer-readable media **2020** in a server computer while another portion of program instructions **2018** can be located in computer-readable media **2020** located in a set of client computers.

(179) The different components illustrated for data processing system **2000** are not meant to provide architectural limitations to the manner in which different embodiments can be implemented. In some illustrative examples, one or more of the components may be incorporated in

or otherwise form a portion of, another component. For example, memory **2006**, or portions thereof, may be incorporated in processor unit **2004** in some illustrative examples. The different illustrative embodiments can be implemented in a data processing system including components in addition to or in place of those illustrated for data processing system **2000**. Other components shown in FIG. **20** can be varied from the illustrative examples shown. The different embodiments can be implemented using any hardware device or system capable of running program instructions **2018**.

(180) Illustrative embodiments of the disclosure may be described in the context of aircraft manufacturing and service method **2100** as shown in FIG. **21** and aircraft **2200** as shown in FIG. **22**. Turning first to FIG. **21**, an illustration of an aircraft manufacturing and service method is depicted in accordance with an illustrative embodiment. During pre-production, aircraft manufacturing and service method **2100** may include specification and design **2102** of aircraft **2200** in FIG. **22** and material procurement **2104**.

(181) During production, component and subassembly manufacturing **2106** and system integration **2108** of aircraft **2200** in FIG. **22** takes place. Thereafter, aircraft **2200** in FIG. **22** can go through certification and delivery **2110** in order to be placed in service **2112**. While in service **2112** by a customer, aircraft **2200** in FIG. **22** is scheduled for routine maintenance and service **2114**, which may include modification, reconfiguration, refurbishment, and other maintenance or service.

(182) Each of the processes of aircraft manufacturing and service method **2100** may be performed or carried out by a system integrator, a third party, an operator, or some combination thereof. In these examples, the operator may be a customer. For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors, and suppliers; and an operator may be an airline, a leasing company, a military entity, a service organization, and so on.

(183) With reference now to FIG. **22**, an illustration of an aircraft is depicted in which an illustrative embodiment may be implemented. In this example, aircraft **2200** is produced by aircraft manufacturing and service method **2100** in FIG. **21** and may include airframe **2202** with plurality of systems **2204** and interior **2206**. Examples of systems **2204** include one or more of propulsion system **2208**, electrical system **2210**, hydraulic system **2212**, and environmental system **2214**. Any number of other systems may be included. Although an aerospace example is shown, different illustrative embodiments may be applied to other industries, such as the automotive industry.

(184) Apparatuses and methods embodied herein may be employed during at least one of the stages of aircraft manufacturing and service method **2100** in FIG. **21**.

(185) In one illustrative example, components or subassemblies produced in component and subassembly manufacturing **2106** in FIG. **21** can be fabricated or manufactured in a manner similar to components or subassemblies produced while aircraft **2200** is in service **2112** in FIG. **21**. As yet another example, one or more apparatus embodiments, method embodiments, or a combination thereof can be utilized during production stages, such as component and subassembly manufacturing **2106** and system integration **2108** in FIG. **21**. One or more apparatus embodiments, method embodiments, or a combination thereof may be utilized while aircraft **2200** is in service **2112**, during maintenance and service **2114** in FIG. **21**, or both. The use of a number of the different illustrative embodiments may substantially expedite the assembly of aircraft **2200**, reduce the cost of aircraft **2200**, or both expedite the assembly of aircraft **2200** and reduce the cost of aircraft **2200**.

(186) For example, in one or more illustrative examples, maintenance manager **214** in FIG. **2** can reduce cost and time associated with pulling and replacing parts in maintenance and service **2114**. Further, increased availability of aircraft during in service **2112** and reduction of time of aircraft in maintenance and service **2114** can occur through improved procedures in pulling and replacing parts based on the alerts generated by maintenance manager **214**. In other words, actions can be

taken to update maintenance procedures as to what parts to pull and replace. Further, actions can also be taken to adjust testing procedures for parts that have been pulled for testing.

(187) Thus, the illustrative embodiments provide a method, apparatus, and computer program product for managing maintenance for vehicles such as aircraft. A computer system determines whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type. The no fault found parameter is how often a no fault found for the part type occurs relative to a number of times that the part type has been removed and replaced. The computer system determines whether an average fix time parameter is greater than a fix time threshold for a correct part replacement. The average fix time parameter identifies an average amount of time before a replaced part for the part type was replaced again. The computer system generates an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than fix time threshold for the correct part replacement.

(188) In the illustrative examples, the analysis of information regarding no fault found parts can be used to reduce cost and time in performing maintenance for aircraft. The illustrative examples can determine that a no fault found part may have been pulled unnecessarily as part of a maintenance procedure. The illustrative examples can also determine that a testing procedure resulting in finding a part is no fault found may be incorrect or lacking to accurately identify an issue with the part. Either or both of these situations can be present based on the alerts generated in the different illustrative examples. As a result, these alerts can be used to perform further investigations or analysis of at least one of maintenance procedures or testing procedures. The result of these investigations analysis can be the creation of at least one of a new maintenance procedure or a new testing procedure that reduces potential issues with having too many no fault found parts in inventory as spare parts.

(189) The description of the different illustrative embodiments has been presented for purposes of illustration and description and is not intended to be exhaustive or limited to the embodiments in the form disclosed. The different illustrative examples describe components that perform actions or operations. In an illustrative embodiment, a component can be configured to perform the action or operation described. For example, the component can have a configuration or design for a structure that provides the component an ability to perform the action or operation that is described in the illustrative examples as being performed by the component. Further, to the extent that terms “includes”, “including”, “has”, “contains”, and variants thereof are used herein, such terms are intended to be inclusive in a manner similar to the term “comprises” as an open transition word without precluding any additional or other elements.

(190) Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other desirable embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A method for managing aircraft parts, the method comprising: receiving, by a computer system from a flight management system of an aircraft, a fault indicator for a part type of the aircraft; determining, by the computer system, whether a no fault found parameter for the part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type, wherein the no fault found parameter is how often a no fault found for the part type occurs relative to a number of times that the part type has been removed and replaced; determining,

by the computer system, whether an average fix time parameter is greater than a fix time threshold for a correct part replacement, wherein the average fix time parameter identifies an average amount of time before a replaced part for the part type was replaced again; generating, by the computer system, an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than fix time threshold for the correct part replacement; displaying, by the computer system, part types in a graphical user interface on a display system in the computer system; and displaying, by the computer system, an alert graphical indicator in association with the part type in the graphical user interface, wherein the alert graphic indicator visually indicates the alert in the graphical user interface.

2. The method of claim 1 further comprising: determining, by the computer system, whether a fix count identifying a number of times removal and replacement of the aircraft parts for the part type fixed a problem is greater than a median fix count multiplied with a fix count median threshold; wherein generating the alert comprises: generating, by the computer system, the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for a correct part replacement; and the fix count identifying the number of times removal and replacement of aircraft parts for the part type fixed the problem being greater than the median fix count multiplied with the fix count median threshold.

3. The method of claim 1 further comprising: determining, by the computer system, whether a shop count identifying a number of times the aircraft parts for the part type have been removed and replaced being greater than a median shop count multiplied with a shop count median threshold; wherein generating the alert comprises: generating, by the computer system, the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for the correct part replacement; and shop count identifying a number of times aircraft parts for the part type have been removed and replaced being greater than a median shop count multiplied with a shop count median threshold.

4. The method of claim 3 further comprising: determining, by the computer system, whether a fix count identifying a number of times removal and replacement of the aircraft parts for the part type fixed a problem is greater than a median fix count multiplied with a fix count median threshold; determining, by the computer system, whether a shop count identifying a number of times aircraft parts for the part type have been removed and replaced is greater than a median shop count multiplied with a shop count median threshold; wherein generating the alert comprises: generating, by the computer system, the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for the correct part replacement; the fix count identifying the number of times removal and replacement of aircraft parts for the part type fixed the problem is greater than the median fix count multiplied with the fix count median threshold; and the shop count identifying the number of times aircraft parts for the part type have been removed and replaced being greater than the median shop count multiplied with a shop count median threshold.

5. The method of claim 1 further comprising: displaying, by the computer system, the part types in rows in the graphical user interface; and displaying, by the computer system, at least one of a shop count, a no fault found count, average fix time, or no fault found percent in columns in the graphical user interface.

6. The method of claim 1 further comprising: displaying, by the computer system, a graph with at least one of a shop count, a no fault found count, average fix time, or a no fault found percent over selected periods of time in the graphical user interface in response to a selection of the part type.

7. The method of claim 1 further comprising: displaying, by the computer system, a no fault found

percent in the graphical user interface using a no fault found graphical indicator comprising a number representing the no fault found percent and a trend line representing a trend for the no fault found percentage.

8. The method of claim 7 further comprising: displaying, by the computer system, a percent change in the trend over a selected period of time in the graphical user interface in response to a selection of the no fault found graphical indicator.

9. An aircraft maintenance system comprising: a computer system; and a maintenance manager in the computer system, wherein the maintenance manager is configured to: determine whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type, wherein the no fault found parameter is how often a no fault found for the part type occurs relative to a number of times the part type has been removed and replaced; determine whether an average fix time parameter is greater than a fix time threshold for a correct part replacement, wherein the average fix time parameter identifies an average amount of time before a replaced aircraft part for the part type was replaced again; generate an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than the fix time threshold for the correct part replacement; display part types in a graphical user interface on a display system in the computer system; and display an alert graphical indicator in association with the part type in the graphical user interface, wherein the alert graphic indicator visually indicates the alert in the graphical user interface.

10. The aircraft maintenance system of claim 9, wherein the maintenance manager is configured to: determine whether a fix count identifying a number of times removal and replacement of the aircraft parts for the part type fixed a problem is greater than a median fix count multiplied with a fix count median threshold; wherein generating the alert comprises: generate the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for a correct part replacement; and the fix count identifying the number of times removal and replacement of aircraft parts for the part type fixed the problem being greater than the median fix count multiplied with the fix count median threshold.

11. The aircraft maintenance system of claim 9, wherein the maintenance manager is configured to: determine whether a shop count identifying a number of times the aircraft parts for the part type have been removed and replaced is greater than a median shop count multiplied with a shop count median threshold; wherein in generating the alert, the maintenance manager is configured to: generate the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for the correct part replacement; and shop count identifying a number of times aircraft parts for the part type have been removed and replaced being greater than a median shop count multiplied with a shop count median threshold.

12. The aircraft maintenance system of claim 11, wherein the maintenance manager is configured to: determine whether a fix count identifying a number of times removal and replacement of the aircraft parts for the part type fixed a problem is greater than a median fix count multiplied with a fix count median threshold; determine whether a shop count identifying a number of times aircraft parts for the part type have been removed and replaced is greater than a median shop count multiplied with a shop count median threshold; wherein in generating the alert, the maintenance manager is configured to: generate the alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for the no fault found occurrences for the part type; the average fix time parameter being greater than the fix time threshold for the correct part replacement; the fix count identifying the number of times removal

and replacement of aircraft parts for the part type fixed the problem is greater than the median fix count multiplied with the fix count median threshold; and the shop count identifying the number of times aircraft parts for the part type have been removed and replaced being greater than the median shop count multiplied with a shop count median threshold.

13. The aircraft maintenance system of claim 9, wherein the maintenance manager is configured to: display the part types in rows in the graphical user interface; and display at least one of a shop count, a no fault found count, average fix time, or no fault found percent in columns in the graphical user interface.

14. The aircraft maintenance system of claim 9, wherein the maintenance manager is configured to: display a graph with at least one of at least one of a shop count, a no fault found count, average fix time, or a no fault found percent over selected periods of time in the graphical user interface in response to a selection of the part type.

15. The aircraft maintenance system of claim 9, wherein the maintenance manager is configured to: display a no fault found percent in the graphical user interface using a no fault found graphical indicator comprising a number representing the no fault found percent and a trend line representing a trend for the no fault found percentage.

16. The aircraft maintenance system of claim 15, wherein the maintenance manager is configured to: display a percent change in the trend over a selected period of time in the graphical user interface in response to a selection of the no fault found graphical indicator.

17. A computer program product for managing aircraft parts, the computer program product comprising a computer readable storage media having program instructions embodied therewith, the program instructions executable by a computer system to cause the computer system to perform a method of: determining whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type, wherein the no fault found parameter is how often a no fault found for the part type occurs relative to a number of times the part type has been removed and replaced; determining whether an average fix time parameter is greater than a fix time threshold for a correct part replacement, wherein the average fix time parameter identifies an average amount of time before a replaced part for the part type was replaced again; generating an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than the fix time threshold for the correct part replacement; display part types in a graphical user interface on a display system in the computer system; and display an alert graphical indicator in association with the part type in the graphical user interface, wherein the alert graphic indicator visually indicates the alert in the graphical user interface.

18. A vehicle maintenance system comprising: a computer system; and a maintenance manager in the computer system, wherein the maintenance manager is configured to: determine whether a no fault found parameter for a part type is greater than a no fault found threshold for an acceptable level for no fault found occurrences for the part type, wherein the no fault found parameter is how often a no fault found for the part type occurs relative to a number of times the part type has been removed and replaced; determine whether an average fix time parameter is greater than a fix time threshold for a correct part replacement, wherein the average fix time parameter identifies an average amount of time before a replaced vehicle part for the part type was replaced again; generate an alert in response to the no fault found parameter for the part type being greater than the no fault found threshold for the acceptable level for no fault found occurrences for the part type and the average fix time parameter being greater than the fix time threshold for the correct part replacement; display part types in a graphical user interface on a display system in the computer system; and display an alert graphical indicator in association with the part type in the graphical user interface, wherein the alert graphic indicator visually indicates the alert in the graphical user interface.

